# Adaptive Sampling for Efficient Multi-Scale Network Centrality Analysis

Author Name
*Affiliation*
email@example.com

February 9, 2026

**Abstract**

Localised network centrality, computed across multiple distance thresholds, characterises accessibility and flow in street networks but is effectively quadratic in network size. Source-sampling bounds such as Eppstein–Wang adapt to the localised setting by replacing total node count $n$ with distance-bounded reach $r$. The resulting Hoeffding model, $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1,\, k/r)$, has zero fitted parameters ($\varepsilon$ and $\delta$ are user-chosen conventions). We show that at $\varepsilon = 0.1$ this model delivers both formal additive-error guarantees and rank preservation (Spearman $\rho \geq 0.98$) on real street networks. Validation on the Greater London (294,000 nodes) and Greater Madrid networks at distances from 1–20 km confirms this, with speedups of 5.1–63.2×. The approach is implemented in the open-source `cityseer` Python package.

**Keywords:** network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

## 1 Introduction

Urban morphological analysis makes use of localised network centrality (typically variants of closeness or betweenness computed within distance thresholds) to reveal patterns of accessibility and flow in street networks. Because these metrics are typically evaluated at multiple scales, from pedestrian (∼500 m) to metropolitan (∼20 km), the computational cost compounds: each distance threshold requires a shortest-path computation from every node, and the reachable subgraph grows with the threshold. For a network of 294,000 nodes, a single multi-scale centrality analysis can take hours.

Sampling-based approximation, in which shortest paths are computed from a random subset of source nodes and the results rescaled, can reduce this cost substantially. A body of theoretical work provides formal sample complexity bounds for global centrality on general graphs, expressed in terms of total node count, graph diameter, VC-dimension, or estimator variance (Eppstein and Wang, 2004; Bader et al., 2007; Riondato and Kornaropoulos, 2016; Borassi and Natale, 2019; Pellegrina and Vandin, 2023). These bounds guarantee uniform additive error $\varepsilon$ over all nodes. For source-sampling approaches such as Eppstein–Wang, adapting the bound to localised centrality is conceptually straightforward: one replaces the global node count $n$ with the distance-bounded reach $r$, since the relevant graph for each distance threshold is the local subgraph of reachable nodes. The resulting localised bound, $O(\log r/\varepsilon^2)$, is valid (Section 3.2). The common assumption, however, is that such Hoeffding-style bounds are far too conservative for practical use. We show that this assumption is incorrect for localised centrality on street networks: at $\varepsilon = 0.1$, the adapted bound prescribes sample sizes that deliver practical speedups while preserving rank order ($\rho \geq 0.98$).

Our contributions are:

1. We adapt the Eppstein–Wang source-sampling bound to localised centrality by replacing global node count $n$ with distance-bounded reach $r$, yielding a zero-parameter Hoeffding model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$. At low reach, $k > r$ and $p = 1$ (exact computation); at high reach, speedup grows as $O(r/\log r)$.

2. We demonstrate that at $\varepsilon = 0.1$, the model delivers both additive-error guarantees and rank preservation ($\rho \geq 0.98$) on real street networks, without any fitted parameters.

3. We validate on the Greater London (294,000 nodes) and Greater Madrid networks at distances from 1–20 km, achieving speedups of 5.1–63.2×.

The approach is implemented in the open-source `cityseer` Python package.

Throughout, we use $r$ to denote the mean reach (average number of nodes reachable from a source within the distance threshold). This scalar predictor governs the per-distance sampling probability.

## 1.1 Computational Cost of Network Centrality

Standard algorithms for localised centrality are built on repeated single-source shortest path computations, but the precise complexity depends on the metric and whether edges are weighted. For distance-based (weighted) closeness variants, one typically runs Dijkstra from every source: each run is $O(M \log N)$ with a binary heap, yielding $O(NM \log N)$ overall. For betweenness, Brandes' algorithm also performs a shortest-path computation from each source followed by a dependency-accumulation pass; the classic bounds are $O(NM)$ for unweighted graphs (BFS per source) and $O(NM + N^2 \log N)$ for weighted graphs (Brandes, 2001). In sparse street graphs where $M = O(N)$, these costs are commonly superlinear and often effectively quadratic in practice.

The practical consequence is straightforward. A network of 294,000 nodes requires 294,000 Dijkstra runs. Even with distance cutoffs that limit individual traversals, the per-source cost grows with the analysis distance: at a 20 km threshold in a dense urban network, each run may visit tens of thousands of nodes. A single centrality computation at one distance can take hours. Urban morphology studies routinely require multiple distance thresholds, typically spanning two orders of magnitude from pedestrian to metropolitan scales, multiplying the total cost accordingly.

## 1.2 Sampling-Based Approximation

Both closeness and betweenness can be expressed as sums over contributions from individual source nodes: each source launches a shortest-path computation that contributes to the centrality of every node it reaches. This additive structure makes sampling straightforward — rather than computing from every source, one can compute from a random subset and rescale to estimate the full sum. Under uniform source sampling, Horvitz–Thompson-style rescaling (dividing by the inclusion probability) yields an unbiased estimator of the population sum; ratio estimators such as Hájek are generally consistent and often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). In either case, sampling error decreases at an approximate $1/\sqrt{k}$ rate with the number of sampled sources, up to finite-population corrections.

Brandes and Pich (2007) introduced this idea for both closeness and betweenness, showing that random pivot selection from a small fraction of source nodes produces useful centrality estimates on networks of tens of thousands of nodes. However, their work did not provide formal error bounds or guidance on choosing the number of sources.

Eppstein and Wang (2004) gave the first formal bound for source-sampled closeness centrality, showing that $O(\log n/\varepsilon^2)$ single-source shortest path computations suffice to approximate the

closeness of all vertices to within additive error $\varepsilon \cdot \Delta_G$ with high probability, where $\Delta_G$ is the graph diameter. Their approach (sampling source nodes uniformly and averaging the resulting distance estimates) is the most directly comparable to ours, as both methods reduce computation by selecting a random subset of sources. The $\log n$ factor in their bound depends on the total number of nodes, regardless of the analysis distance.

Bader et al. (2007) gave the first probabilistic error bound for sampled betweenness, showing that $O(1/\varepsilon^2)$ samples suffice to approximate the betweenness of a single vertex to within additive error $\varepsilon$ with high probability. Their adaptive algorithm doubles the sample count until the confidence interval contracts below a user-specified threshold. The bound applies per vertex and does not directly provide uniform approximation across all vertices simultaneously.

Riondato and Kornaropoulos (2016) addressed the uniform approximation problem using Vapnik–Chervonenkis (VC) dimension theory (Vapnik and Chervonenkis, 1971). They showed that the number of samples required for a uniform additive $\varepsilon$-approximation of all betweenness values depends on the VC-dimension of the set of shortest paths, which is bounded by $\lfloor \log_2(\mathrm{VD}-2) \rfloor + 1$, where VD is the vertex-diameter of the graph. This yields sample complexity logarithmic in the vertex-diameter rather than the number of nodes. The resulting bounds are worst-case, however, and can be conservative for well-structured networks such as urban street grids.

For closeness centrality specifically, Cohen et al. (2014) proposed an approximation framework that exploits the structure of closeness (a sum of inverse distances) to achieve tighter bounds than generic betweenness methods. Their approach uses random sampling of target nodes rather than source nodes and provides per-vertex confidence intervals.

Borassi and Natale (2019) introduced KADABRA, an adaptive algorithm for betweenness that adjusts the number of samples online. KADABRA maintains running confidence intervals for each vertex's betweenness and terminates when all intervals are sufficiently narrow, eliminating the need to specify the number of samples in advance. Bergamini et al. (2019) provided a comparative evaluation of these and related approaches, finding that adaptive methods generally outperform fixed-sample methods in terms of accuracy per sample, though the overhead of online monitoring can offset some of the savings.

Most recently, Pellegrina and Vandin (2023) introduced SILVAN, which replaces VC-dimension bounds with Monte Carlo Rademacher averages to obtain tighter, *variance-dependent* sample complexity bounds for betweenness. Because SILVAN's bounds depend on the actual variance of the betweenness estimator rather than worst-case combinatorial quantities, they can be substantially tighter on well-structured graphs. However, SILVAN still targets global betweenness with additive $\varepsilon$-guarantees.

These methods divide into two fundamentally different sampling strategies. *Source-sampling* (Brandes, Eppstein–Wang, Bader) selects a random subset of source nodes and runs a full shortest-path tree from each; every node reachable from a sampled source receives a centrality contribution. *Path-sampling* (Riondato–Kornaropoulos, KADABRA, SILVAN) instead draws random source–target pairs and computes a single shortest path between them, counting betweenness contributions along that path. Cohen's approach is a third variant, *target-sampling*, which samples destination nodes and propagates distance information backward. The distinction matters because standard centrality algorithms, including localised implementations with distance cutoffs, are inherently source-based: each source launches a Dijkstra traversal that is naturally bounded by the distance threshold $d_{\max}$, producing contributions to every node within reach. Source-sampling bounds therefore analyse the same computational primitive that localised centrality uses, whereas path-sampling bounds analyse a different primitive whose relationship to distance-bounded computation is less straightforward.

A common feature of this body of work is that it targets *global* centrality, computed over the entire graph without distance cutoffs, on general (non-spatial) networks. Urban network analysis, by contrast, uses *localised* centrality with distance thresholds that range from hundreds of metres to tens of kilometres. The reachable subgraph at each threshold varies by orders of

magnitude, from hundreds of nodes at pedestrian scales to tens of thousands at metropolitan scales. For source-sampling bounds such as Eppstein–Wang, adapting to the localised setting is straightforward: the relevant graph has $r$ nodes rather than $n$, and the bound follows by substitution. Whether the resulting sample sizes are practical, rather than merely valid, is the question we address.

# 2 Methods

## 2.1 Centrality Measures

We consider two centrality measures, both computed with a distance cutoff $d_{\max}$ as is standard in urban network analysis. **Harmonic closeness** (Rochat, 2009; Freeman, 1979) sums inverse distances to all nodes within the threshold:

$$C_H(v;\, d_{\max}) = \sum_{u:\, d(v,u) \le d_{\max}} \frac{1}{d(v,u)} \tag{1}$$

**Betweenness** (Freeman, 1977) counts the fraction of shortest paths (with total length at most $d_{\max}$) that pass through each node:

$$C_B(v;\, d_{\max}) = \sum_{\substack{s \ne v \ne t \\ d(s,t) \le d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{2}$$

where $\sigma_{st}$ is the number of shortest paths from $s$ to $t$, and $\sigma_{st}(v)$ is the number passing through $v$. Common distance thresholds range from 400 m to 20 km.

A quantity central to the sampling analysis is *reach*: the mean number of nodes within $d_{\max}$,

$$r = \frac{1}{N} \sum_{v} |\{u : d(v,u) \le d_{\max}\}| \tag{3}$$

In the Greater London network, reach ranges from 2,783 at 5 km to 43,165 at 20 km. This order-of-magnitude variation across distance thresholds directly governs the relationship between sample size and accuracy.

## 2.2 Experimental Design

Our analysis proceeds in three stages: (1) empirical characterisation of sampling accuracy on synthetic networks with controlled topology, (2) derivation of the Hoeffding sampling model from the Eppstein–Wang bound, and (3) validation on large real-world street networks.

We generate synthetic networks using 3 topology templates: trellis (high connectivity, representing planned urban cores), tree (low connectivity, representing suburban sprawl), and linear (intermediate connectivity). These span the range of urban street network structure. Each topology is tiled to create networks of approximately 12km extent, of which we use the inner 4 square. We test 7 distance thresholds from 250m to 4,000m, and 12 sampling probabilities from 2.5% to 100%.

A critical methodological consideration is the treatment of boundary effects. Nodes near network edges have artificially lower centrality because their catchments are truncated. We address this by applying an inward buffer from the convex hull of all nodes, equal to the maximum analysis distance (4km for synthetic networks, 20km for validation). Only nodes inside this buffered zone are marked as "live" and included in accuracy calculations, ensuring all live nodes have complete catchments at all tested distances. This approach concurs with best practice for localised network analysis (Gil, 2017).

### 2.3 Accuracy Metrics

We measure sampling accuracy using two complementary metrics: Spearman rank correlation and normalised additive error.

**Spearman rank correlation** ($\rho$) between sampled and ground-truth centrality values:

$$\rho = 1 - \frac{6 \sum_i (R_i^{\text{sampled}} - R_i^{\text{true}})^2}{N(N^2 - 1)} \tag{4}$$

where $R_i$ denotes the rank of node $i$. Spearman correlation is appropriate because urban applications typically care about *relative* rankings of streets rather than absolute centrality values. More fundamentally, Spearman $\rho$ is structurally aligned with the accuracy profile that source-sampling creates. Under sampling, each node's estimate depends on how many sampled sources contribute to it, which is proportional to its reachability. High-reachability nodes, which are also the high-centrality nodes, receive contributions from many sampled sources and have precise estimates; low-reachability peripheral nodes receive fewer contributions and have noisier estimates. Spearman $\rho$ naturally tolerates imprecision where centrality values are tightly clustered (rank swaps among near-equal peripheral nodes have negligible effect on $\rho$) while being sensitive to misrankings among the well-separated high-centrality nodes where sampling is most precise.

**Normalised additive error** $\varepsilon_{\text{obs}} = \max_v |C^{\text{sampled}}(v) - C^{\text{true}}(v)| \,/\, C_{\max}$, where $C_{\max}$ is the theoretical maximum contribution of each source: $r(r-1)$ for betweenness and $r$ for harmonic closeness. The resulting normalised error $\varepsilon \in [0, 1]$ is directly comparable to the $\varepsilon$ parameter in the Hoeffding bound and serves as the formal accuracy measure.

### 2.4 Validation Networks

We validate the model on two real-world street networks. The primary validation uses the Greater London Authority (GLA) road network, sourced from Ordnance Survey Open Roads data. After preprocessing (removing filler nodes, consolidating nearby nodes, and removing dangles), the network comprises approximately 294,000 nodes. We apply a 20km inward buffer to eliminate edge artifacts. As an external validation, we also test the model on the Greater Madrid metropolitan street network following the same protocol.

## 3 The Hoeffding Sampling Model

### 3.1 The Opportunity: Sampling Can Work

Figure 1 illustrates why sampling-based centrality is attractive. Panel A shows that accuracy (Spearman $\rho$) increases with sampling probability across all distance thresholds, but the curves flatten well before 100% sampling. At 4000m distance, $\rho \geq 0.95$ is achieved at just 10% sampling, a 10× potential speedup. Panel B shows that, for a given level of accuracy, the relative speedup from reduced sampling becomes more pronounced as the analysis distance grows. The remainder of this section derives a sampling model from the Eppstein–Wang bound that prescribes sampling probability as a function of reach alone.

### 3.2 Adapting the Eppstein–Wang Bound to Localised Centrality

The Eppstein–Wang framework (Eppstein and Wang, 2004) bounds the error of source-sampled centrality using Hoeffding's inequality (Hoeffding, 1963). When $k$ sources are sampled uniformly, each sampled source contributes a bounded random variable to each target node's normalised centrality estimate. Hoeffding's inequality, combined with a union bound over all target nodes, gives a probability guarantee that the maximum normalised error across all nodes is at most $\varepsilon$. In the original global setting, the union bound is over $n$ nodes, yielding sample complexity $k = \log(2n/\delta)/(2\varepsilon^2)$.

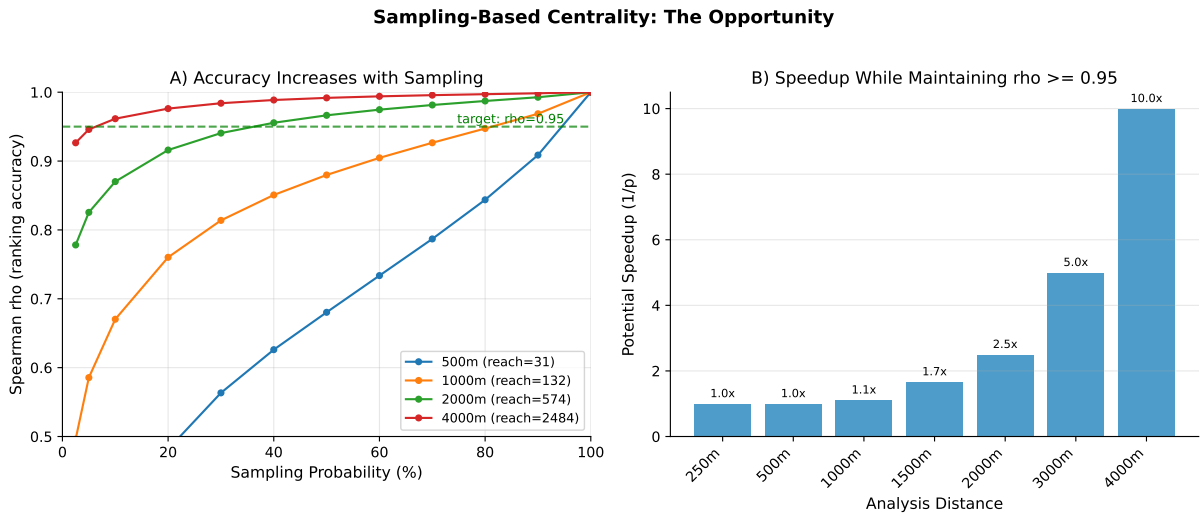


Figure 1: **The sampling opportunity.** (A) Accuracy vs. sampling probability for different analysis distances. The green dashed line marks the $\rho = 0.95$ target. (B) Potential speedup at each distance while maintaining $\rho \geq 0.95$.

For localised centrality with distance threshold $d_{\max}$, the relevant graph at each distance has $r$ reachable nodes rather than $n$ total nodes. The adaptation is direct: substituting $r$ for $n$ yields

$$k = \frac{\log(2r/\delta)}{2\varepsilon^2} \tag{5}$$

with corresponding sampling probability

$$p = \min\left(1,\ \frac{k}{r}\right) \tag{6}$$

This model has zero fitted parameters: $\varepsilon$ (error tolerance) and $\delta$ (failure probability) are user-chosen conventions. At default values $\varepsilon = 0.1$, $\delta = 0.1$:

- **Low reach** ($r \lesssim 500$): $k > r$, so $p = 1$ and computation is exact. No separate floor parameter is needed; the natural saturation of $p$ at 1 handles small networks automatically.
- **High reach**: $p$ decreases as $O(\log r/r)$, and the theoretical speedup $1/p$ grows as $O(r/\log r)$. At reach 10,000, $p \approx 6.1\%$ (16.4× speedup); at reach 50,000, $p \approx 1.4\%$ (72× speedup).

Figure 2 shows the model's prescribed effective sample size ($k$) and speedup ($1/p$) as functions of reach for three $\varepsilon$ values. The logarithmic growth of $k$ with $r$ means that the sampling probability decreases rapidly with reach, producing speedups that grow with network size.

Table 1 provides the required effective sample sizes and sampling probabilities at three $\varepsilon$ values across a range of reach values.

### 3.3 Error Structure Across Reachability

Although the Hoeffding bound guarantees uniform additive error, it is natural to ask how accuracy varies across the reachability distribution. We pool all nodes across all synthetic configurations at the Hoeffding-prescribed sampling probability ($\varepsilon = 0.1$) and bin them by absolute per-node reach. Figure 3 reveals a crossover: absolute error increases with reach, while normalised error decreases. This pattern has a natural interpretation. High-reach nodes sit at the centres of large local subgraphs; their centrality values are large sums over many source

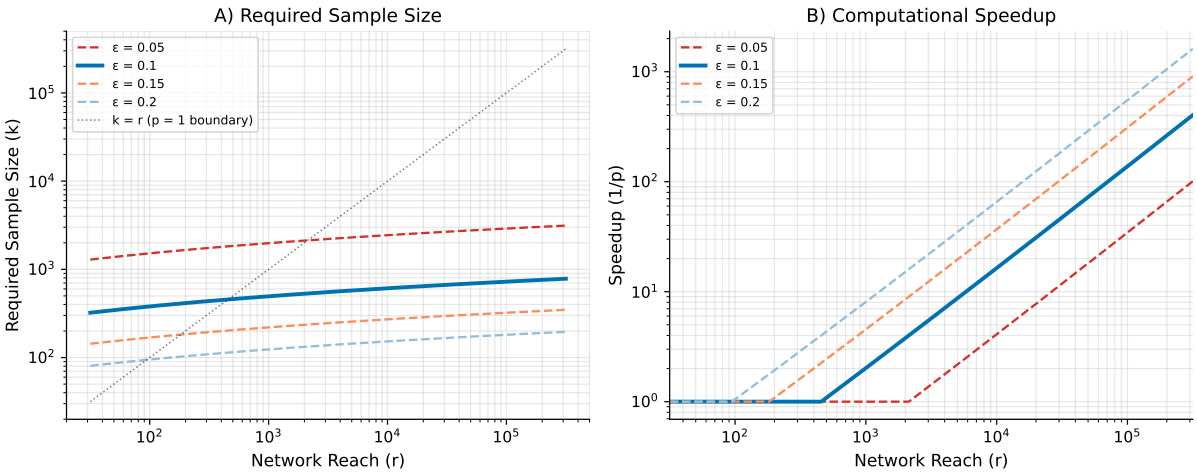


Figure 2: **The Hoeffding sampling model.** (A) Required effective sample size $k$ vs. reach for three error tolerances ($\varepsilon = 0.05, 0.1, 0.2$). The $\log r$ dependence produces slowly growing sample requirements. (B) Theoretical speedup $1/p$ vs. reach (log-log scale). At the default $\varepsilon = 0.1$, speedup exceeds $10\times$ beyond reach $\approx 10{,}000$.

Table 1: Required effective sample sizes and sampling probabilities under the localised Eppstein–Wang bound at different additive error tolerances ($\delta = 0.1$).

| **Reach** | **EW** $\varepsilon$=0.05 | $p$ | **EW** $\varepsilon$=0.1 | $p$ | **EW** $\varepsilon$=0.2 | $p$ |
|---|---|---|---|---|---|---|
| 100 | 1,520 | 100.0% | 380 | 100.0% | 95 | 95.0% |
| 500 | 1,842 | 100.0% | 460 | 92.1% | 115 | 23.0% |
| 1,000 | 1,980 | 100.0% | 495 | 49.5% | 123 | 12.4% |
| 3,000 | 2,200 | 73.3% | 550 | 18.3% | 137 | 4.6% |
| 10,000 | 2,441 | 24.4% | 610 | 6.1% | 152 | 1.5% |
| 30,000 | 2,660 | 8.9% | 665 | 2.2% | 166 | 0.6% |

The Hoeffding model prescribes $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$. At the default $\varepsilon = 0.1$, $\delta = 0.1$, the model has zero fitted parameters.

contributions, so absolute estimation errors accumulate. But these same nodes enjoy high effective sample sizes ($n_{\text{eff}} = p \cdot r$), so the errors are small relative to the centrality magnitude. Conversely, low-reach peripheral nodes have small absolute errors but larger relative errors. In practical terms, the nodes that matter most for urban analysis (high-centrality arterials and junctions) receive the highest precision, a property we term *precision scales with importance*.

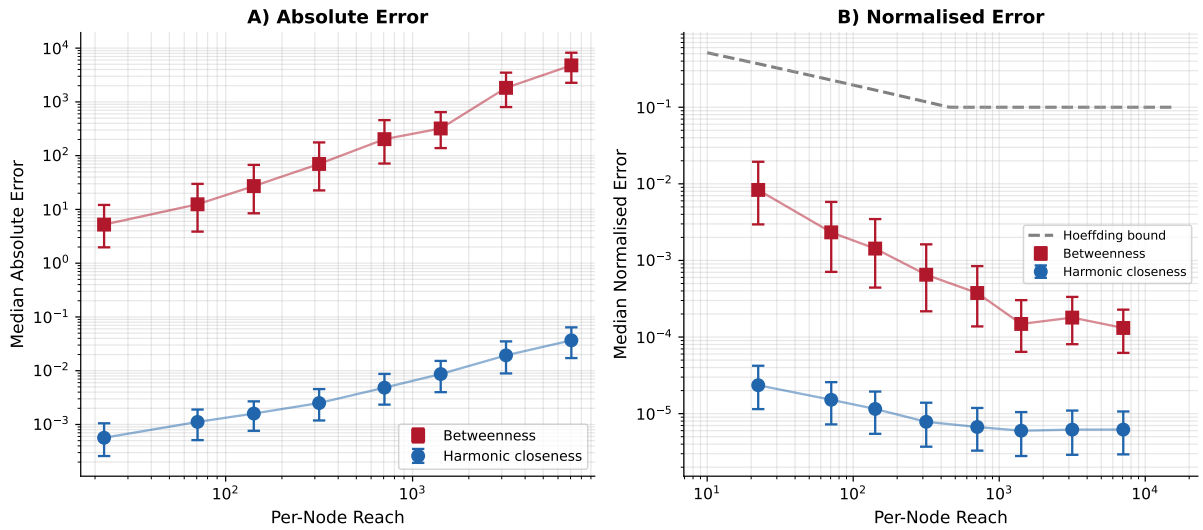


Figure 3: **Error crossover: precision scales with importance.** Per-node errors pooled across all synthetic configurations at the Hoeffding-prescribed sampling probability ($\varepsilon = 0.1$), binned by absolute per-node reach (log scale). Error bars show interquartile range across nodes within each bin. (A) Absolute error increases with reach: high-centrality nodes accumulate larger absolute errors because their centrality values are large sums over many source contributions. (B) Normalised error decreases with reach: relative error shrinks because high-reach nodes enjoy higher effective sample sizes ($n_{\text{eff}} = p \cdot r$). The dashed grey line shows the Hoeffding bound ($\varepsilon = \sqrt{\log(2r/\delta)/(2 \cdot n_{\text{eff}})}$, $\delta = 0.1$): observed median errors are two to three orders of magnitude below the worst-case prediction. Normalisers are $r$ for closeness and $r(r-1)$ for betweenness.

This error structure explains why additive $\varepsilon = 0.1$ implies strong rank preservation ($\rho \geq 0.98$). High-reach nodes, which are precisely the high-centrality nodes whose ranking matters most, get the highest effective sample sizes and the most precise estimates. Low-reach peripheral nodes get near-exact computation ($p \to 1$). The only nodes with appreciable normalised error are at intermediate reach, where centrality values are tightly clustered and rank swaps have negligible effect on Spearman $\rho$.

### 3.4 Relationship to Prior Bounds

Table 2 places our approach in the context of the existing sampling literature.

Among the bounds surveyed here, Eppstein–Wang is the only one that analyses source-sampling, the same computational primitive our algorithm uses. The adaptation to localised centrality is direct substitution ($r$ for $n$). Path-sampling bounds (Riondato–Kornaropoulos, KADABRA, SILVAN) do not adapt as directly, since they rely on global graph primitives (vertex diameter, VC-dimension) and a sampling primitive different from distance-bounded Dijkstra. Bader et al.'s per-vertex bound does not directly provide simultaneous guarantees.

The key observation is that on street networks, the adapted bound at $\varepsilon = 0.1$ is not merely valid but *practical*: observed errors are two to three orders of magnitude below the worst-case prediction (Figure 3B). This conservatism means that the bound delivers far better accuracy

Table 2: Comparison of sampling approaches for centrality approximation. All prior methods target global centrality (no distance cutoff) on general graphs. Our approach adapts the Eppstein–Wang bound to localised centrality.

| **Method** | **Metric** | **Error type** | **Sample complexity** | **Key parameter** |
|---|---|---|---|---|
| Eppstein–Wang | Closeness | Additive $\varepsilon$ | $O(\log n \,/\, \varepsilon^2)$ | Node count $n$ |
| Bader et al. | Betweenness | Additive $\varepsilon$ | $O(1 \,/\, \varepsilon^2)$ | Per vertex |
| Riondato–Korn. | Betweenness | Uniform $\varepsilon$ | $O(\mathrm{VD} \,/\, \varepsilon^2)$ | VC-dimension |
| KADABRA | Betweenness | Uniform $\varepsilon$ | $O(\log D \,/\, \varepsilon^2)$ | Diameter $D$ |
| SILVAN | Betweenness | Non-uniform $\varepsilon$ | Variance-dependent | Path variance |
| **This work** | **Both** | **Additive** $\varepsilon$ | $\boldsymbol{O(\log r \,/\, \varepsilon^2)}$ | **Reach** $r$ |

than it guarantees, making $\varepsilon = 0.1$ a pragmatic default that yields both formal error guarantees and strong rank preservation.

### 3.5 Adaptive Sampling Algorithm

The model determines a sampling probability for each distance threshold based on reach. Because reach varies with distance, optimal sampling also varies: short distances (low reach) may require full computation, while long distances enable aggressive sampling. Algorithm 1 presents the complete procedure.

**Algorithm 1** Adaptive Per-Distance Sampling for Network Centrality

**Require:** Network $G = (V, E)$, distance thresholds $D = \{d_1, \ldots, d_m\}$
**Require:** Error tolerance $\varepsilon$ (default 0.1), confidence $\delta$ (default 0.1)
**Ensure:** Centrality estimates $\hat{C}[d]$ for each $d \in D$

```
Phase 1: Reach Estimation
for each distance d ∈ D do
    Compute reach r_d ← mean nodes within distance d
end for
Phase 2: Adaptive Sampling
for each distance d ∈ D do
    Compute required sources: k_d ← log(2r_d/δ)/(2ε²)
    Compute sampling probability: p_d ← min(1, k_d/r_d)
    Sample sources S_d ⊆ V with probability p_d
    Compute centrality from S_d: Ĉ[d] ← Centrality(G, S_d, d)
    Scale estimates: Ĉ[d] ← Ĉ[d]/p_d
end for
return {Ĉ[d] : d ∈ D}
```

The reach $r_d$ is typically a byproduct of the centrality computation itself; when a quick estimate is needed beforehand, a pilot sample of $\approx 100$ nodes distributed on a square kilometre grid suffices. Sources are sampled uniformly at random, and sampled values are scaled by $1/p$ to produce unbiased estimates.

With sampling probability $p$, we compute from $pN$ sources instead of $N$, reducing single-source computations by a factor of $1/p$. For a network with distances spanning 500 m to 20 km, the model prescribes $p = 1.0$ at 500 m (reach $\approx 200$, no benefit), $p \approx 11.5\%$ at 5 km (reach $\approx 3{,}000$), and $p \approx 3.2\%$ at 20 km (reach $\approx 40{,}000$), yielding a theoretical $31.0\times$ speedup at the longest distance.

# 4 Validation

## 4.1 Experimental Protocol

We validate the model on the GLA and Greater Madrid networks described in Section 2. For the GLA network, we compute ground-truth centrality ($p = 1.0$) and sampled centrality at the Hoeffding-prescribed probability ($\varepsilon = 0.1$, $\delta = 0.1$) at each distance threshold, measuring Spearman $\rho$ between sampled and ground-truth values. For the Madrid network, we follow the same protocol, repeating each configuration with 5 random seeds and reporting the mean $\pm$ standard deviation to assess replication stability.

## 4.2 GLA Results

Table 3 presents validation results for the Greater London network at five distances from 1–20 km. At all distances, both harmonic closeness and betweenness achieve $\rho > 0.95$, with the minimum observed $\rho = 0.98$ (betweenness, 20km). At short distances (1–2 km), the model prescribes $p = 1.0$ (exact computation), confirming that the natural floor behaviour is appropriate. At longer distances, speedups grow from 5.1× at 5 km to 63.2× at 20 km.

Table 3: Hoeffding model validation on Greater London network ($\varepsilon = 0.1$, $\delta = 0.1$).

| **Distance** | **Reach** | **Hoeff.** $p$ | $\rho_{\mathbf{close}}$ | $\rho_{\mathbf{betw}}$ | **Speedup** | $\rho \geq 0.95$**?** |
|---|---|---|---|---|---|---|
| 1km | 100 | 100.0% | 0.9948 | 0.9983 | 1.0× | ✓ |
| 2km | 411 | 100.0% | 0.9979 | 0.9993 | 1.0× | ✓ |
| 5km | 2,783 | 19.6% | 0.9875 | 0.9912 | 5.1× | ✓ |
| 10km | 11,590 | 5.3% | 0.9930 | 0.9846 | 18.8× | ✓ |
| 20km | 43,165 | 1.6% | 0.9959 | 0.9824 | 63.2× | ✓ |

Network: Greater London Area, 294,486 nodes, 20km live node buffer. Hoeffding model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$.

Betweenness is consistently the binding constraint: at every distance, $\rho_{\text{betw}} < \rho_{\text{close}}$. This is expected because betweenness contributions are sparser and heavier-tailed than closeness contributions (most sources route zero shortest paths through a given node, while a few contribute heavily), resulting in higher per-source variance. The Hoeffding bound, which treats both metrics identically, is therefore more conservative for closeness.

## 4.3 External Validation: Madrid

To test generalisability beyond London, we applied the identical protocol to the Greater Madrid metropolitan street network. Table 4 presents the results. The model achieves $\rho > 0.95$ for both closeness and betweenness at all tested distances, confirming that the approach generalises across morphologically distinct European cities. The minimum observed $\rho = 0.9794$ (betweenness, 20km). Speedups range from 2.8× at 5 km to 24.2× at 20 km, lower than GLA because the Madrid network is smaller (fewer reachable nodes at each distance).

# 5 Discussion and Conclusion

## 5.1 Practical Guidelines

For users of the `cityseer` library, the Hoeffding sampling model is applied automatically. For other implementations, the procedure is: (1) estimate reach $r$ at each analysis distance, (2) compute $k = \log(2r/\delta)/(2\varepsilon^2)$ and $p = \min(1, k/r)$ with default $\varepsilon = 0.1$, $\delta = 0.1$, and (3) compute centrality from a random $p$-fraction of sources, scaling results by $1/p$.

Table 4: Hoeffding model validation on Greater Madrid network ($\varepsilon = 0.1$, $\delta = 0.1$).

| **Distance** | **Reach** | **Hoeff.** $p$ | $\rho_{\mathbf{close}}$ | $\rho_{\mathbf{betw}}$ | **Speedup** | $\rho \geq 0.95$**?** |
|---|---|---|---|---|---|---|
| 5km | 1,451 | 35.4% | 0.9969 | 0.9949 | 2.8× | ✓ |
| 10km | 5,219 | 11.1% | 0.9974 | 0.9872 | 9.0× | ✓ |
| 20km | 15,300 | 4.1% | 0.9976 | 0.9794 | 24.2× | ✓ |

Network: Greater Madrid, 99,166 nodes, 20km live node buffer. Hoeffding model: $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$.

Figure 4 provides visual lookup charts for common scenarios, and Table 5 gives a numerical reference for representative reach values.

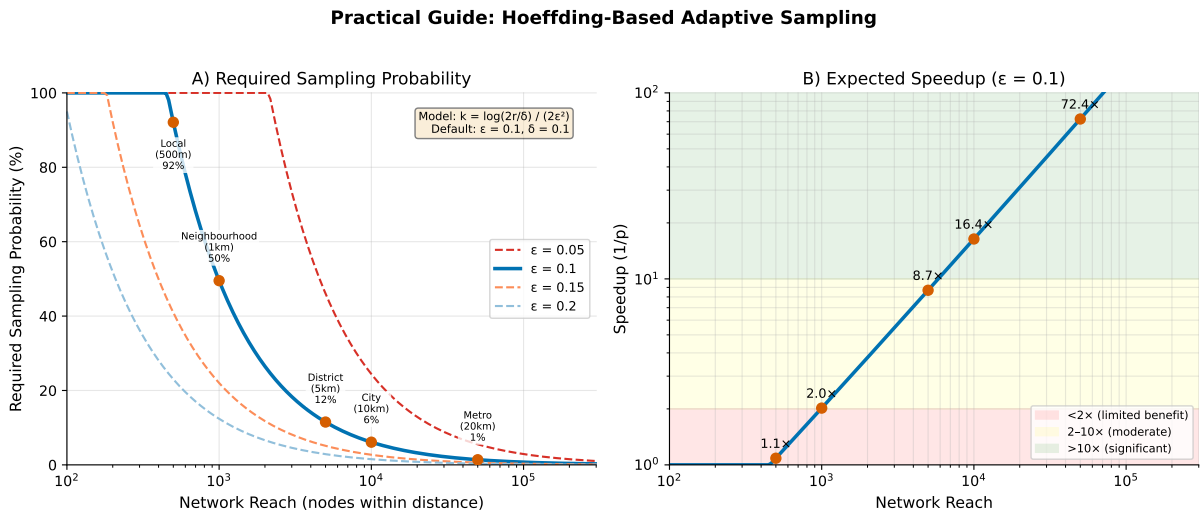


Figure 4: **Practical guidance.** (A) Required sampling probability by network reach, with common urban scenarios annotated. (B) Expected computational speedup.

As a rough guide: at reach below ∼500, $p = 1.0$ (exact computation). At reach 1,000–10,000, sample 5–50% (2–16× speedup). At reach above 10,000, sample below 6% (> 16× speedup). Users requiring stricter guarantees can reduce $\varepsilon$ (e.g., $\varepsilon = 0.05$ roughly quadruples the required samples) or average 2–3 replications with different random seeds.

Sampling is not appropriate when exact centrality values (not approximations) are required, when centrality is needed for specific nodes rather than the full network, or when the network is small enough (< 1,000 nodes) that full computation is already fast.

## 5.2 Limitations

Our approach rests on several assumptions. First, we assume uniform random sampling of source nodes. Non-uniform strategies (stratified by location or weighted toward high-centrality sources) could potentially achieve higher accuracy with fewer samples, but would require information about node importance that is unavailable before computation. Second, the model treats the network as statistically homogeneous: accuracy depends on global reach, not local structure. Our validation on the GLA and Madrid networks, which exhibit heterogeneous topology (dense urban cores surrounded by sparse suburbs), shows the model remains conservative overall, but accuracy may vary locally. Third, our experiments used shortest-path distance; angular (simplest-path) distance, common in space syntax analysis, may exhibit different accuracy characteristics and warrants separate validation. Fourth, validation is limited to two European cities; networks with substantially different morphology (e.g., North American grids, informal settlements) would

Table 5: Practical sampling lookup table ($\varepsilon = 0.1$, $\delta = 0.1$).

| **Reach** | $k$ | $p$ (%) | **Speedup** |
|---|---|---|---|
| 100 | 380 | 100.0 | 1.0× |
| 200 | 415 | 100.0 | 1.0× |
| 500 | 461 | 92.1 | 1.1× |
| 1,000 | 495 | 49.5 | 2.0× |
| 2,000 | 530 | 26.5 | 3.8× |
| 5,000 | 576 | 11.5 | 8.7× |
| 10,000 | 610 | 6.1 | 16.4× |
| 20,000 | 645 | 3.2 | 31.0× |
| 50,000 | 691 | 1.4 | 72.4× |
| 100,000 | 725 | 0.7 | 137.8× |

**Usage:** Find your network reach (nodes within analysis distance), read off the required sampling probability.
**Model:** $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1,\ k/r)$. At low reach, $k > r$ so $p = 1$ (exact computation).

strengthen the generality claim. Finally, betweenness shows thinner margins than closeness at high reach, and the minimum observed $\rho = 0.9794$ (Madrid betweenness, 20km), while above the 0.95 threshold, leaves less headroom than closeness.

### 5.3 Future Directions

Several extensions could build on this work. Adaptive per-node sampling, where the probability varies with local reach, could improve efficiency in heterogeneous networks. Importance sampling, weighting toward high-centrality sources, could reduce variance. Online stopping criteria, as in KADABRA (Borassi and Natale, 2019), could replace a priori calibration with adaptive termination based on observed convergence, but adapting such methods to localised, distance-bounded centrality would require new theoretical development. Systematic validation for angular distance and extension to other centrality measures (eigenvector centrality, PageRank) would broaden applicability. SILVAN's variance-dependent bounds (Pellegrina and Vandin, 2023), which replace worst-case combinatorial quantities with the actual estimator variance, represent a promising direction for tightening the localised bound on well-structured network classes.

We have shown that the Eppstein–Wang source-sampling bound, adapted to localised centrality by replacing global node count with distance-bounded reach, yields a practical zero-parameter sampling model. The Hoeffding model $k = \log(2r/\delta)/(2\varepsilon^2)$, $p = \min(1, k/r)$ at $\varepsilon = 0.1$ delivers both formal additive-error guarantees and strong rank preservation ($\rho \geq 0.98$) on real street networks. Rank preservation arises structurally: high-reach nodes get high effective sample sizes and precise estimates; these are precisely the high-centrality nodes whose ranking matters. Low-reach peripheral nodes get near-exact computation ($p \to 1$). Validation on the Greater London (294,000 nodes) and Greater Madrid networks at 1–20 km confirms speedups of 5.1–63.2× with all configurations exceeding $\rho \geq 0.95$.

The model is implemented in the open-source `cityseer` Python library.[1] Analysis scripts and validation data are available in the repository's `analysis/sampling/` directory.

## References

Bader, D. A., Kintali, S., Madduri, K., and Mihail, M. (2007). Approximating Betweenness Centrality. In *Algorithms and Models for the Web-Graph (WAW 2007)*, volume 4863 of

[1] `https://github.com/benchmark-urbanism/cityseer-api`

*Lecture Notes in Computer Science*, pages 124–137. Springer.

Bergamini, E., Borassi, M., Crescenzi, P., Marino, A., and Meyerhenke, H. (2019). Comparing the speed and accuracy of approaches to betweenness centrality approximation. *Computational Social Networks*, 6(1):5.

Borassi, M. and Natale, E. (2019). KADABRA is an ADaptive Algorithm for Betweenness via Random Approximation. In *27th Annual European Symposium on Algorithms (ESA 2019)*, pages 20:1–20:18.

Brandes, U. (2001). A faster algorithm for betweenness centrality. *Journal of Mathematical Sociology*, 25(2):163–177.

Brandes, U. and Pich, C. (2007). Centrality Estimation in Large Networks. In *Proceedings of the 33rd International Workshop on Graph-Theoretic Concepts in Computer Science (WG 2007)*, volume 4769 of *Lecture Notes in Computer Science*, pages 90–100. Springer.

Cochran, W. G. (1977). *Sampling Techniques*. Wiley, 3rd edition.

Cohen, E., Delling, D., Pajor, T., and Werneck, R. F. (2014). Computing Classic Closeness Centrality, at Scale. In *Proceedings of the Second ACM Conference on Online Social Networks (COSN)*, pages 37–50.

Eppstein, D. and Wang, J. (2004). Fast Approximation of Centrality. *Journal of Graph Algorithms and Applications*, 8(1):27–34.

Freeman, L. C. (1977). A Set of Measures of Centrality Based on Betweenness. *Sociometry*, 40(1):35–41.

Freeman, L. C. (1979). Centrality in Social Networks: Conceptual Clarification. *Social Networks*, 1(3):215–239.

Gil, J. (2017). Street network analysis "edge effects": Examining the sensitivity of centrality measures to boundary conditions. *Environment and Planning B: Urban Analytics and City Science*, 44(5):819–836.

Hagberg, A. A., Schult, D. A., and Swart, P. J. (2008). Exploring Network Structure, Dynamics, and Function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pages 11–15.

Hájek, J. (1971). Comment on "An Essay on the Logical Foundations of Survey Sampling, Part One". *Metrika*, 17(1):1–30.

Hoeffding, W. (1963). Probability Inequalities for Sums of Bounded Random Variables. *Journal of the American Statistical Association*, 58(301):13–30.

Horvitz, D. G. and Thompson, D. J. (1952). A Generalization of Sampling Without Replacement from a Finite Universe. *Journal of the American Statistical Association*, 47(260):663–685.

Pellegrina, L. and Vandin, F. (2023). SILVAN: Estimating Betweenness Centralities with Progressive Sampling and Non-uniform Rademacher Bounds. *ACM Transactions on Knowledge Discovery from Data*, 18(3):1–40.

Riondato, M. and Kornaropoulos, E. M. (2016). Fast Approximation of Betweenness Centrality through Sampling. *Data Mining and Knowledge Discovery*, 30(2):438–475.

Rochat, Y. (2009). Closeness Centrality Extended to Unconnected Graphs: The Harmonic Centrality Index. In *Applications of Social Network Analysis (ASNA 2009)*.

Särndal, C.-E., Swensson, B., and Wretman, J. (2003). *Model Assisted Survey Sampling.* Springer, 2nd edition.

Vapnik, V. N. and Chervonenkis, A. Y. (1971). On the uniform convergence of relative frequencies of events to their probabilities. *Theory of Probability and Its Applications*, 16(2):264–280.

# A Technical Details

## A.1 Synthetic Network Generation

Synthetic networks are generated using tile-based templates. Each template defines a connectivity pattern within a single tile; tiles are then arrayed to create networks of the desired extent.

**Trellis** A regular grid with diagonal connections, representing highly connected urban cores. Average degree $\approx 4$.

**Tree** A hierarchical branching structure where each node connects to at most 3 others. Represents suburban cul-de-sac development. Average degree $\approx 2.3$.

**Linear** A constrained grid with fewer cross-connections than trellis. Intermediate connectivity. Average degree $\approx 3$.

Networks are tiled 24×24 to achieve approximately 12km extent, then the largest connected component is extracted. Node positions are perturbed slightly to avoid perfectly regular spacing.

## A.2 GLA Network Preprocessing

The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that don't represent junctions)
5. Remove dangling nodes (dead-end segments $< 20$m)
6. Consolidate nearby nodes (threshold 1m)
7. Extract largest connected component

Final network: $\approx$ 294,000 nodes.

## A.3 Correctness Verification Against NetworkX

To ensure the correctness of our Rust-based centrality implementation in `cityseer`, we conducted systematic verification against the widely-used NetworkX library (Hagberg et al., 2008). For each synthetic network topology and distance threshold combination, we computed both harmonic closeness and betweenness centrality using our Rust implementation (via `cityseer` Python bindings) and NetworkX's reference functions.

For each node $v$ in the test networks, we computed the absolute difference $\Delta_v = |C_{\text{cityseer}}(v) - C_{\text{NetworkX}}(v)|$. The maximum difference across all nodes and all test configurations was $\max_v \Delta_v < 10^{-6}$, attributable to floating-point arithmetic precision.

Verification was performed across all three synthetic topologies, distance thresholds from 200m to 3200m, both shortest-path and angular distance heuristics, and both harmonic closeness and betweenness centrality.

## A.4 Horvitz–Thompson Estimator

When estimating population totals from samples, two classical approaches are the Horvitz–Thompson (H–T) estimator and the Hájek ratio estimator. Under Bernoulli (independent) source sampling with inclusion probabilities $\pi_i$, the Horvitz–Thompson estimator (Horvitz and Thompson, 1952) estimates a population total as $\hat{T}_{HT} = \sum_{i \in S} y_i/\pi_i$, where $S$ is the set of sampled sources. This inverse-probability weighting yields an unbiased estimator whenever $\pi_i > 0$ for all sampled units.

The Hájek estimator (Hájek, 1971) instead uses a ratio form,

$$\hat{T}_H = N \cdot \frac{\sum_{i \in S} y_i/\pi_i}{\sum_{i \in S} 1/\pi_i}, \tag{7}$$

which is often lower-variance but can be slightly biased in finite samples (Cochran, 1977; Särndal et al., 2003). In `cityseer`, we use Horvitz–Thompson-style inverse-probability weighting because it matches the paper's $1/p$ rescaling description and extends directly to unequal inclusion probabilities (e.g., when combining a base sampling probability with per-source importance weights).

## A.5 Implementation in `cityseer`

The sampling model is implemented in the `cityseer` Python library (version 4.23+). The adaptive function automatically probes reachability, computes sampling probabilities from the Hoeffding model, and scales results:

```
from cityseer.metrics import networks

gdf = networks.node_centrality_shortest_adaptive(
    network_structure,
    nodes_gdf,
    distances=[500, 1000, 2000, 5000, 10000],
    compute_closeness=True,
    compute_betweenness=True,
    epsilon=0.1,
    delta=0.1
)
```

For manual control, a fixed sampling probability can be specified via the `sample_probability` parameter of `node_centrality_shortest`. Helper functions for custom workflows (`compute_hoeffding_p`, `compute_sample_probs`, `probe_reachability`, `spatial_sample`) are available in the `cityseer.config` module; see the library documentation for details.

## A.6 Reproducibility

All analysis scripts and validation results are available at `https://github.com/benchmark-urbanism/cityseer`. The script pipeline runs sequentially from `00_generate_cache.py` through `07_hoeffding_model_figure.py`, with each script's outputs documented in its header.